AJAY GARG

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EDUCATION

B.Tech, Computer Science & Engineering (Cloud Computing) *2023 – 2027*
SRM Institute of Science and Technology, Ghaziabad | CGPA: 8.58/10

EXPERIENCE

Associate Software Developer Intern, NKRA (NKRA_Webapp) *Oct 2026 – Nov 2026*
- Built "Ask NKRA", a customer-facing AI chat assistant for the company website, end to end as the sole developer in JavaScript, in under a month.
- Designed the conversation flow to move visitors from product questions to a hand-off form that routes leads to a human agent.
- Developed in a test environment with synthetic data, following company confidentiality and security requirements.

PROJECTS

TapRemit — Trust-first payments for AI agents (Monad testnet)
Express · TypeScript · SQLite · Zod · Vitest · Next.js 14
- Treated the AI agent as an untrusted proposal generator: a deterministic policy engine (asset whitelist, per-tx and rolling 24h limits, first-time-recipient quarantine) gates every transfer.
- Bound each approval to a SHA-256 hash of the proposal behind a compare-and-swap state machine to block double-approval; logged every step in a hash-chained append-only audit log enforced by SQLite triggers.
- Built a testnet executor with mutex-serialized nonces, write-ahead submission, finality checks and fail-closed handling of stuck transactions; 87 Vitest tests passing.

Hindsight — AI on-call crew with temporal graph memory (FalkorDB hackathon)
Python · FastAPI · FalkorDB · MCP · SSE · Docker Compose
- Built a multi-agent incident-response system on FalkorDB graph memory that detects runbooks invalidated by architecture changes and refuses stale ones; a "Graph OFF" baseline measures the graph's value.
- Coordinated 5 agent workers with atomic Cypher claim/release queries instead of application locks (0 double-claims at 20–32 concurrent threads in tests); blast-radius rules halt risky actions for operator approval.
- Streamed a live "war-room" UI over SSE from FastAPI with an MCP tool registry; deployed publicly with Docker Compose.

VEDA — Agentic AI platform (deployed on AWS EC2; since shut down) *From Jan 2025*
Python · FastAPI · FAISS · Groq/Llama 3.3 70B · Docker · Nginx · GitHub Actions
- Built a multi-agent orchestrator that schedules tasks with Kahn's topological sort over a dependency graph, calling Groq-hosted Llama 3.3 70B with FAISS-backed long-term memory.
- Split it into 4 FastAPI microservices behind an Nginx reverse proxy with JWT middleware; GitHub Actions built and pushed Docker images, with Prometheus/Grafana monitoring.

Digital Vault — Secure vault with video/audio calling *Mar 2024 – Aug 2024*
FastAPI · React/TypeScript · PostgreSQL · Agora · Railway · Vercel
- Designed stateless JWT auth (15-min access, 7-day single-use SHA-256-hashed refresh tokens) to limit token-replay risk; added TOTP 2FA with a pre-auth token pattern and lockout after 5 failed attempts.
- Built 12+ REST endpoints and Agora-based calling; deployed on Railway and Vercel.

HACKATHONS

GreenGate + Phoenix, GitLab Transcend: carbon-aware release agent with Lambda canary and auto-rollback *(Python, GitLab Duo flows, AWS Lambda, EventBridge, S3)*
RXCLEAR, Bharat Builds Tour (WeMakeDevs × AWS): prescription decoder and substitute-medicine finder *(React/Vite/TypeScript, AWS Lambda, API Gateway, S3)*
Threshold, Amazon Developer Hackathon, Ring track: doorbell interpreter for deaf/hard-of-hearing and language-gap residents *(Next.js, Python, AWS)*

TECHNICAL SKILLS

Languages: Python, Java, TypeScript/JavaScript, Go, SQL
Backend / Frontend: FastAPI, Express, REST, SSE, WebSockets, JWT/TOTP auth; React, Next.js, Tailwind
Data / AI: PostgreSQL, MySQL, SQLite, FalkorDB, FAISS, Qdrant, LangChain, LLM integration (Groq, OpenAI, Bedrock), MCP
Cloud / DevOps: AWS (EC2, Lambda, S3, API Gateway, DynamoDB, Bedrock), Docker, Docker Compose, Nginx, GitHub Actions, Prometheus, Grafana
DSA: Java; LeetCode practice

RESEARCH & CERTIFICATIONS

- Co-author, submitted paper: autoencoder anomaly detection and automated remediation for self-healing cloud infrastructure.
- NPTEL Elite, Natural Language Processing, IIT Kharagpur (Jan – Apr 2026)